MBA 753: Causal Inference Models - Assignment 2

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Q1. Conduct an event study analysis to study the change of trend in petrol prices after the Indian government introduced day-to-day changes in fuel prices on 16th June 2017.

a) What is the regression equation for your model?

$$rate = 38.27 + 0.00687(Time) - 2.709(PostEvent) + 0.000247(TimeAfterEvent)$$

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b) What is the relationship of time with petrol prices?

Before the intorduction of day-to-day chanegs of fuel prices, time had a positive and statistically significant **(p<2e-16)** relation with petrol prices. For every additional day, the price of petrol increases by approximately **0.00687** units on average.

After the policy introduction, the change in the trend **(0.00025)** was statistically insignificant **(p=0.314)**. This means the policy did not have significant impact on petrol prices, and they continued to rise at the rate of approximately **0.0071** units per day **(0.00687 + 0.00025)**.

c) What is your conclusion about the policy implemented from 16th June 2017.

The introduction of day-to-day changes in fuel prices on 16th June 2017 resulted in an immediate, significant drop in petrol prices by **2.709** units **(p-value = 1.11e-14)**. However, the policy had no significant long-term effect on the trend of petrol prices; the coefficient for TimeAfterEvent (**0.000247**) is not statistically significant **(p-value = 0.314)**.

Q2. For the event study model above, include state as one of the covariates.

a) What is the regression equation for your model?

$$rate = 35.07 + 0.00686(Time) - 2.689(PostEvent) + 0.000258(TimeAfterEvent) + 2.502(Karnataka)$$
$$+6.562(Maharashtra) + 2.798(TamilNadu) + 4.289(Telangana)$$

b) What is the relationship of time with petrol prices? Did the relationship change from the first model to the second model?

Before the introduction of day-to-day changes in fuel prices, time had a positive and statistically significant **(p<2e-16)** relation with petrol prices. For every additional day, the price of petrol increased by approximately **0.00686** units on average.

After the policy introduction, the *change* in the trend (**0.00026**) was statistically insignificant **(p=0.248)**. This means the policy did not meaningfully alter the trajectory of petrol prices, and they continued to rise at the combined baseline rate of approximately **0.00712** units per day (0.00686 + 0.00026).

The relationship practically did not change between the two models; the **Time** coefficient shifted marginally from **0.00687** to **0.00686**, indicating the baseline trend is robust even when these new state variables are introduced.

c) What can you say about the relationship between states and petrol prices? Is the relationship causal – why or why not?

The state coefficients indicate that Karnataka, Maharashtra, Tamil Nadu, and Telangana all have significantly higher baseline petrol prices compared to the reference state (Delhi). Maharashtra has the highest premium at 6.56 units above the baseline.

The relationship is not causal because a state itself does not inherently cause price differences, the state variable acts as a proxy for unobserved, time-invariant, state-level differences such as local transportation costs, taxes (VAT), and regional supply chains.

Q3. Fit two event studies model – one for Karnataka and one for Maharashtra.

a) Write the regression equations of model for both the states.

Karnataka:

$$rate = 97.317 - 0.006(Time) + 3.554(PostEvent) + 0.014(TimeAfterEvent)$$

Maharashtra:

$$rate = 38.114 + 0.008(Time) - 1.234(PostEvent) - 0.002(TimeAfterEvent)$$

b) Compare and interpret the causal relationship of policy implementation for the two states.

- **Post Change**: In Karnataka, the policy caused an immediate and significant increase in prices by **3.554** units. Conversely, Maharashtra experienced an immediate decrease in prices by **1.234** units.
- **Trend Change**: Before the policy, Karnataka had a negative trend of **-0.006**, but the policy shifted the trend upward significantly by **+0.014 per day**. Maharashtra had a positive trend of **+0.008**, and the policy shifted this trend slightly downward by **-0.002 per day**.
- **Causal Relation**: The policy implementation had different effects on the two trajectories. It completely reversed Karnataka's negative trend, driving prices up over time. Conversely, Maharashtra's trend did not reverse; its already positive price growth was flattened slightly by the policy.

References

[1] Cunningham, S. (2021). Causal Inference: The Mixtape. Yale University Press.